



Deep learning-based automated identification on vortex-induced vibration of long suspenders for the suspension bridge

Xun Su^a, Jianxiao Mao^{a,*}, Hao Wang^{a,*}, Hui Gao^a, Dan Li^a

^a Key Laboratory of Concrete and Prestressed Concrete Structures of Ministry of Education, Southeast University, Nanjing 211189, China

ARTICLE INFO

Communicated by: Wei-Xin Ren

Keywords:

Bridge suspenders
vortex-induced vibration (VIV)
Automated identification
Deep learning
Structural health monitoring

ABSTRACT

Large amplitude vortex-induced vibration (VIV) of long suspenders, caused by vortex-shedding and wake flow, will decrease the service life of high-strength wires and its anchorage systems, hence listed as an essential monitoring indicator of the structural health monitoring system in the suspension bridges. Traditionally, a static vibration amplitude limitation is usually predefined and used to identify VIV from continuously monitored acceleration data of bridge suspenders. Due to the complexity of the starting vibration of the suspenders, it may not be able to flexibly adapt to the suspenders VIV under different conditions. In addition, transient but significant vibration events may not be captured since it relies only on static vibration amplitude limits without considering the temporal information of vibration. To address the above-mentioned issues, a novel framework to autonomously identify the suspenders VIV is proposed using the multimodal fusion techniques with deep neural networks. The main contributions of the methodology are: i) by calculating the wind field and vibration characteristic, two vibration response parameters are identified as indexes for the identification of suspenders VIV based on the significant difference method of big data analysis; ii) Two different modal information of time history images and power spectral density (PSD) sequences are used as inputs to the convolutional neural network (CNN) and bidirectional long short-term memory (BiLSTM) to extract the suspenders vibration response features from the time and frequency domain perspectives, respectively; and iii) The multimodal information is concentrated and enhanced by the multi-head attention (MHA) mechanism to achieve automated identification of the suspenders VIV. The proposed framework is validated with data collected from a full-scale long-span suspension bridge in comparison with the base model. The results show that the proposed framework can efficiently identify the full-cycle development process of the suspenders VIV. This study provides a convenient strategy for suspenders VIV identification, which can be used for bridge management and vibration mitigation device design.

1. Introduction

Modern cable-supported bridges with high structural efficiency, strong spanning capacity, aesthetic shape, convenient construction, are one of the main types of bridges across rivers, bays and canyons and other mega-obstacles. Since the 21st century, with the

* Corresponding author.

* Corresponding author.

E-mail addresses: jianxiao@seu.edu.cn (J. Mao), wanghao1980@seu.edu.cn (H. Wang).

development of construction technology and the increasing demand, the span of bridges has been continuously expanding. This has made them more flexible and more susceptible to wind-induced vibrations [1–3]. Vortex-induced vibration (VIV) is a typical wind-induced finite-amplitude vibration caused by the interaction of the structure with the incoming wind, especially for slender and flexible structures or components, such as main girders and suspenders of long-span bridges. VIV has always been a hot topic in bridge engineering [4–7]. Although VIV does not lead to bridge collapse, it can cause serious fatigue issues to critical components of the bridge and driving safety concerns [4]. The monitoring, identifying, and tracking VIV can not only alleviate operational stress of long-span bridges but are also crucial for bridge managers to make decisions, prevent potential accidents, and avert widespread public concern. Meanwhile, it provides evidence for the wind-resistant design, structural health monitoring, and intelligent maintenance of long-span bridges [6–8].

As one of the main load-bearing components of long-span suspension bridges, suspenders are characterized by light self-weight, high flexibility, and low damping. VIV is the typical vibration form they encountered at normal wind speeds. Presently, investigations pertaining to VIV of suspenders primarily concentrate on numerical simulation and wind tunnel test. Computational fluid dynamics numerical simulation is prevalent strategy for examining the intricate behaviors of VIV. The approach reveals a tri-dimensional exposition of essential characteristics including vortex magnitude, pressure profiles, and velocity vectors, thereby enabling prognostic insights into both the genesis and the evolution of vortex phenomena [9–12]. Conventional simulation approaches encompass Large Eddy Simulation (LES), Reynolds-Averaged Navier-Stokes (RANS), and Direct Numerical Simulation (DNS). The efficacy of numerical simulations is circumscribed by factors such as computational precision, computational expenditure, and inherent uncertainties [13]. Wind tunnel experiments furnish empirical data concerning actual flow fields, which may be employed to corroborate the outcomes of numerical simulations, and they serve as a customary modality in the investigation of VIV [14–18]. Furthermore, some scholars have conducted research on VIV based on empirical and semi-empirical models [19–25]. They are incapable of accounting for the intricacies and complexities of flow fields. It is necessary to integrate numerical simulations with experimental methodologies, continuously refining and enhancing the accuracy of predictions. During long term service, variations in environmental and operational conditions (e.g., temperature in the bridge site area, traffic flow, wind fields, and other factors) can significantly affect the propensity for VIV [26]. Structural health monitoring (SHM) is the most direct means of obtaining structural response. The reasons for the unpredictability mentioned above emphasize the importance of continuous monitoring of the suspenders VIV to accurately assess vibration.

The widespread implementation of SHM systems, combined with the rapid advancement of machine learning (ML) and deep learning (DL) technologies, has rendered data-driven approaches feasible for VIV identification [27,28]. Data-driven models, often perceived as obscure black boxes, lack transparent physical explications, merely necessitating the transformation of inputs into outputs through an affine relationship. Nevertheless, the implicit mapping relationship between the inputs and outputs of data-driven models may encapsulate physical characteristics pertinent to VIV, which is essential for comprehending the VIV phenomena [29]. Furthermore, encouraged by the increasing availability of low-cost sensors, multimodal fusion using data obtained from different sources and structures for classification or regression provides a new idea for machine learning. It has shown significant advantages over unimodal paradigms in semantic segmentation, action recognition, visual quizzing, and many others, through the introduction of end-to-end neural integration of multimodality.

A novel multimodal identification model for fusing multi-channel image sequences is developed in this study to automatically identify the whole development process of the suspenders VIV. The goal is to exploit the potential of the CNN block to extract complex features from the input image, as well as the ability of the BiLSTM block to discover correlations between sequences through a twofold understanding of the context in each case. To evaluate the performance of the architecture, it was applied to a long-span suspension bridge and compared to four baseline networks to determine if it is worth using a hybrid model.

1.1. Contributions

1. A novel CNN-BiLSTM-MHA multimodal classification framework was developed for suspenders VIV identification.
2. The framework was applied to identify the suspenders VIV of a long-span suspension bridge and performed well in three evaluation metrics.
3. To demonstrate the plausibility of the framework, a comparison was made with several commonly used robust baseline networks.
4. The framework was applied to the parametric analysis of the full process development of several randomly selected VIV events from the field monitoring database.

1.2. Outline

The rest of the paper is organized as follows. In Section 2, the related works is briefly reviewed. In section 3, the selection of the main characteristic indexes utilized in the framework is presented. And the data collection and data pre-processing stages are explained. In Section 4, the proposed multimodal classification framework is introduced. In Section 5, the validity and rationality of the proposed framework is verified in comparison with baseline networks. In Section 6, the conclusions and future work are drawn.

2. Related works

In the face of voluminous monitoring data, discerning the quintessential characteristics of VIV and harnessing data mining techniques to unearth novel insights constitute two pressing challenges awaiting resolution on field monitoring [30–32]. Feature extraction

plays an indispensable role in the precision and generalizability of the model to identify VIV. Currently, the key feature indexes extracted by scholars in the process of identifying and predicting VIV are shown in Table 1. They mainly include: peak factors, kurtosis, acceleration root mean square (RMS), energy concentration coefficient (ECC), wind velocity, wind direction, similarity ratio of amplitude (SRA), and the coefficient of variation (COV). Based on such typical characteristics, the technological frameworks used can be broadly classified into three categories. The first type is unsupervised ML models, which are mainly based on cluster analysis [30–35]. Kim et al. [30] and Li et al. [29] used automated fast clustering algorithms to identify VIV of bridge decks, respectively. He et al. [33] introduced a k-mean clustering method to automatically identify and track VIVs by automatically separating feature indicator vectors into multiple clusters. The second type is to develop a classification model [27,36–38]. Lim et al. [27] used semi-supervised labelling to automatically label wind and vibration feature datasets to train DNN models. Based on the classification method and interaction mechanism of multi-span interaction, Xu et al. [37] investigated the interaction mechanism of multi-span pipelines from both static and dynamic perspectives, and proposed two multi-span classification methods based on modal frequency and energy transfer. Another is based on a regression model for VIVs prediction. Huang et al. [39] proposed a Random Incremental Technique (RDT)-based method that defines a threshold for the COV based on the peak of the processed data to distinguish VIV and non-VIV. Dan et al. [31] proposed a real-time monitoring, intelligent perception and early warning technology for suspension bridge VIV based on recursive Hilbert transform and real-time recursive acceleration integration algorithm. Several researches have compared and combined the advantages of various ML techniques to further improve the performance and identification accuracy of models and frameworks [40–42]. In addition, to capture the nonlinearity in a straightforward manner, a few studies have developed time-domain models to perform VIVs feature extraction [32,43]. These studies provide insights into the development of data-driven predictive models for VIVs and help to improve identification performance. Though these classifiers have achieved impressive performance, there remain the following issues:

- (1) Existing studies are mainly based on the VIV identification of main girders in suspension bridges, and there are few reports on the VIV identification of suspenders.
- (2) Unlike the main girder VIV, the suspenders VIV has more vibration modes and are more readily triggered. The key features extracted in existing studies are not fully applicable for the identification of suspenders VIV.
- (3) Previous investigations have corroborated the identification accuracy of the VIV; discrepancies persist in terms of generalization capabilities and universality when applied across varied long-span bridges.

Considering the multiple parameters that can affect the occurrence of VIVs under uncertainty, this study proposes the multimodal classification framework. A succession of experiments is also conducted to evaluate performance of the framework on different datasets.

3. Engineering background

3.1. Brief description of the target bridge

The Nanjing Qixiashan Yangtze River Bridge (NQYRB) investigated in this study has three spans with lengths of 576.2 m, 1418 m and 481.8 m. The long-span bridge was the first three-span suspension bridge in China and ranked third among similar bridges in the world at that time, as shown in Fig. 1. According to the different load conditions and locations, the bridge is equipped with three types of suspenders: the first type is the displacement-limiting suspender (DLS) located at the top of the transition pier; the second type is the special suspender (SS) which is subjected to higher loads on one side of the pylon and adjacent DLS with special requirements for deformation; and the third type is the ordinary suspender (OS). All of them consist of parallel steel wire bundles, and are connected to the cable clamps and steel box girders by pin hinges. Among them, the long suspenders include two kinds of OS and SS.

A VBM system was built to directly monitor the acceleration of the long suspenders and the surrounding wind field to investigate the properties and excitation mechanisms of the long suspenders VIV. For the accelerometer on the long suspenders, the maximum

Table 1
Summary of research methods for VIV identification.

Types	Authors	Characteristics	Method
Cluster analysis	Hui Li, et al. [32]	Acceleration RMS, peak-to-peak ratio	Fast clustering algorithm
	Arul M et al. [34]	Peak factors, kurtosis, acceleration RMS, ECC	Hierarchical clustering algorithm
Classification model	Peng Liang, et al. [33]	SRA, peak-to-peak ratio	k-mean clustering algorithm
	Hui Li, et al. [29]	Wind velocity, wind direction	Decision tree learning algorithm
	Shiqiao Xu, et al. [36]	Acceleration RMS, peak-to-peak ratio, wind velocity, wind direction	Parameter optimization algorithm
	Sunjoong Kim, et al. [30]	Peak factors, kurtosis, acceleration RMS	Deep neural network
Regression model	Danhui Dan, et al. [31]	SRA	VIV index and real-time analysis method
	Zhiwen Huang, et al. [39]	COV	random decrease technique

Note: SRA = similarity ratio of amplitude; ECC = Energy concentration coefficient; COV = Coefficient of variation

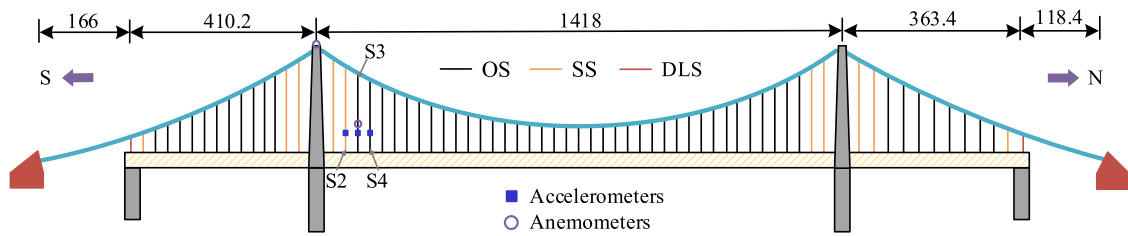


Fig.1. Demonstration of the NQYRB and VBM systems (unit: m).

measuring range is 500 m/s^2 and the sampling frequency is 200 Hz. The ultrasonic anemometer was mounted on a vertical pole at a height of 4 m above the bridge deck surface with a sampling frequency of 10 Hz. All wind data analyzed in this study were obtained from inflow anemometers, which can be considered as natural winds measured without interference from bridge components. The long suspenders investigated are the second, third and fourth sets of suspenders behind the pylons, referred to as S2, S3 and S4. Among them, S3, S4 are OS and S2 is SS. The dimensions and characteristics are shown in Table 2.

To reduce the non-stationary effect of measurement data, 1-minute statistics rather than instantaneous measurements are used for modelling in this study. For the collected data, several faults are first excluded: (1) anomalous values, i.e., 1-minute average wind speeds greater than 50 m/s or RMS of long suspenders vibrations exceeded 500 m/s^2 ; (2) sensor failures, including noise interference, measurement failures and loss of accuracy.

3.2. Analysis of wind speed and direction

The monitoring data of the wind environment in the bridge site area within the last one year are collected. To eliminate the high variability caused by gusts, 1-minute averages are used to represent instantaneous wind speeds. A total of 3420 wind loading conditions for suspenders VIV are investigated. Similarly, 3420 non-VIV wind loads are randomly selected from the bridge monitoring data as the calculation condition. The average wind speeds of the suspenders at VIV and non-VIV and the statistical information are shown in Fig. 2. As shown in Fig. 2(a), the orange scatters are the mean wind speed distributions measured at the occurrence of VIV. Through statistical calculation, the wind speed distribution is found to range from 2.2 m/s $\sim$ 11.2 m/s. The blue scatters are the mean wind speed distribution measured under non-VIV. The wind speed distribution ranges from 0.1 m/s $\sim$ 11.1 m/s. The mean wind speed distribution in the case of VIV has no significant specificity to that of the non-VIV events, except for the wind speed at which vibration begins to develop. It can be deduced that the average wind speed cannot be a characteristic index for determining whether suspenders VIV has occurred.

The wind rose diagram based on wind speed and wind direction data are shown in Fig. 2(b). It can be seen that there is a wide range of wind directions when VIV events occur. Basically, the event of suspenders VIV occurs in the wind direction range on both sides of the bridge axis and is not limited to the direction perpendicular to the bridge axis. Therefore, the wind direction is not exclusive. It also cannot be used as an index to identify VIV. In conclusion, it is difficult to extract effective VIV identifiers for suspenders from wind load excitation. It is necessary to further investigate the vibration characteristics of the suspenders VIV to seek suitable indexes to identify suspenders VIV from the vibration characteristics.

3.3. Variation characteristics of VIV of the bridge suspenders

The early identification and perception of the whole process characteristics of a VIV event as it occurs, including the developmental, stabilization and decline stages, allows bridge managers to have a real-time status of the suspenders. This allows assessment and intervention for the suspenders VIV. However, to effectively identify the occurrence of VIV, it is of utmost importance to extract indicators that reflect the VIV characteristics. It is necessary to analyse the vibration response data of the suspenders to describe the VIV. One of the recorded VIV events of the suspender is presented in Fig. 3. It is clear that the vibration acceleration of the suspender in the initial phase is less than 10 m/s^2 . When VIV occurs, the amplitude of the vibration acceleration of the suspender increases dramatically. From 0.18 h to 0.8 h it increases rapidly to over 80 m/s^2 . After this the vibration is intensive and the vibration acceleration remain around 80 m/s^2 . Until 1.42 h, it began to gradually decline, and the vibration acceleration amplitude gradually decreased to below 10 m/s^2 . This VIV event lasted almost 2 h. From the above analyses and previous studies by scholars [28], it can be clearly found that the vibration characteristics of suspenders under normal conditions are significantly different from those under the action of VIV. The

Table 2
The dimensions and characteristics of investigated suspenders.

Number	Diameter(m)	Tension(kN)	Equivalent volumetric weight (kN/m^3)	Length(m)	Distance(m)	Fundamentalfrequency (Hz)
S2	0.097	1087.25	82.42	145.17	0.58	0.610
S3	0.072	822.77	85.35	138.66	0.48	0.797
S4	0.072	705.8	85.35	132.42	0.48	0.751

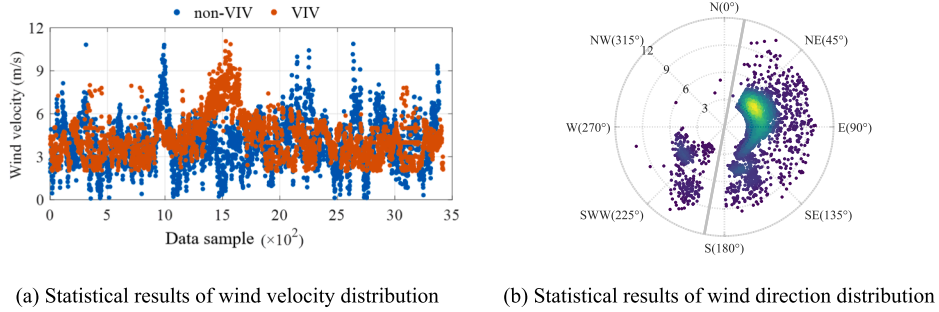


Fig.2. Statistical analysis of wind field monitoring data.

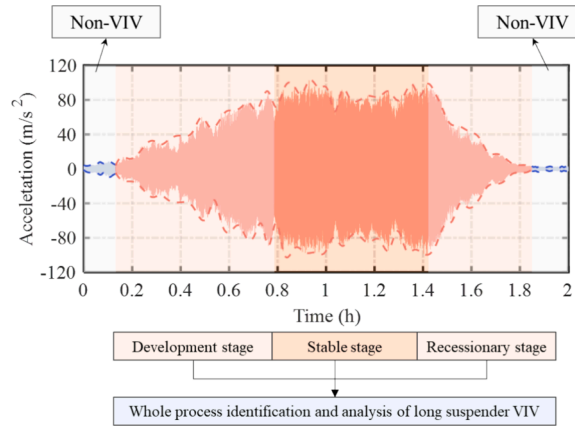


Fig.3. Whole procedure recording of suspenders VIV event.

vibration acceleration of the suspenders during VIV are highly specific and can be used to monitor and identify VIV events based on this characteristic. However, the vibration characteristics of suspenders with different diameters and lengths are not identical, and it is necessary to statistically analyses the vibration characteristics of different suspenders to improve the practicability of the model. Three groups of representative suspenders (one set of special suspenders and two sets of normal suspenders) are investigated.

3.3.1. The acceleration RMS of the bridge suspenders

In this study, the amplitude and root mean square (RMS) values of vibration acceleration are taken as parametric data describing the vibration intensity. To fully grasp the vibration characteristics, the RMS values of the suspender vibration over the last year are statistically analyzed. As can be seen from Fig. 4, the RMS values of the suspenders at VIV are almost always above 5 m/s^2 , and some are even close to 50 m/s^2 . For normal vibration, RMS values vary between $0 \text{ m/s}^2 \sim 10 \text{ m/s}^2$, with most RMS values below 5 m/s^2 . Thus, the RMS values have significant specificity during VIV and can be used as an indicator for modelling the identification of VIVs. When VIV occurs, the RMS value increases rapidly and is greater than a certain threshold. To determine the threshold, the results of RMS values of the suspender during VIV events in the last year are calculated (Fig. 4(b)). The probability density function (PDF) reveals

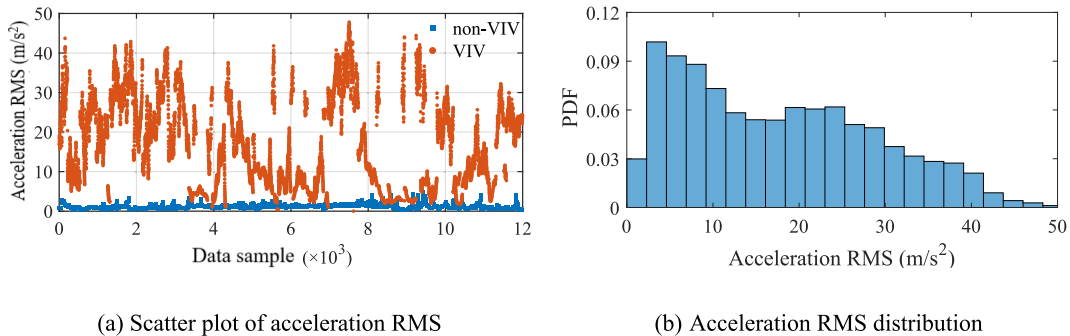


Fig.4. Statistical analysis of suspenders vibration acceleration RMS.

that the RMS values of vibration acceleration when VIV occurs are distributed in the range of $6 \text{ m/s}^2 \sim 50 \text{ m/s}^2$. 87 % of the VIV events have an RMS value greater than 5 m/s^2 . Based on the distribution of the RMS value, a threshold value of 5 m/s^2 is determined. When the RMS value tends to increase over a period of time and is greater than 5 m/s^2 , the suspenders VIV is likely to occur.

3.3.2. The acceleration PSD of the bridge suspenders

To further explore the characteristic vibration parameters of the suspender during VIV, the power spectral density (PSD) analysis of the vibration acceleration is performed. Due to the different wind speeds at different heights of the wind profile and the different vortex shedding frequencies, the vibration modes excited are varied, and the VIV of the suspenders shows the form of multi-modal vibration. Consequently, the suspenders VIV exhibits multi-modal high frequency vibration patterns, with the dominant vibration modes generally being above the 10th order [4,8]. When VIV occurs, the frequency with the maximum energy of PSD is the dominant frequency and can be identified as the mode in which the VIV is activated [31]. The PSD value with the maximum energy is extracted from the power spectrum analysis, as shown in Fig. 5. From the results of the power spectrum analysis, it is clear that the most significant difference between VIV and other vibrations is the frequency energy magnitude. When VIV occurs, the frequency energy distribution is between 1 and 1000, basically above 10, while other vibrations are basically less than 1. This characteristic can be used as an indicator of the occurrence of VIV. The distribution of PSD energy maximums for VIV events and non-VIV events are calculated to obtain an accurate threshold for the indicator. The distribution of PSD energy maximums for VIV events and non-VIV events are calculated to obtain an accurate threshold for the index. It is clear from the histogram that the basic distribution of non-VIV is below 1, while VIV is just the opposite, with most of them located above 1. Those data that do not conform to the above pattern may be due to errors in recording the occurrence time of VIV, resulting in the mixing of VIV with non-VIV events. Therefore, the PSD energy maximum threshold that can be used to determine whether or not VIV has occurred is set to 1. That is, when it is less than 1, the probability of VIV occurring in the suspenders can be considered to be high.

The identification indices of suspender VIV are explored by analyzing the wind field characteristics as well as the amplitude and PSD of the VIV events. Statistical analyses of the wind field characteristics reveal that there is no clear threshold for wind speeds during suspenders VIV and non-VIV, and that there are apparent common intervals. And the wind direction is particularly widely distributed. It is incredibly challenging to identify the suspenders VIV adopting the wind field characteristics as an index. During VIV, the RMS value of the suspender vibration acceleration is basically greater than 5 m/s^2 , and even reaches 50 m/s^2 . Therefore, when the RMS value reaches more than 5 m/s^2 and shows an increasing trend over a period of time, it can be used as an index to predict the occurrence of VIV. The vibration energy maximum can also an index for evaluating VIV in PSD analysis. That is, when it is greater than 1, suspenders are likely to experience VIV.

4. Methodology

4.1. Related theories

4.1.1. Convolutional neural network

CNN represent a class of deep feedforward neural networks. It is constructed by mimicking the connection patterns of neurons in the human visual cortex [44,45]. The typical CNN architecture usually consists of convolutional (Conv) layer, activation layer, pooling (POOL) layer and fully connected (FC) layer as shown in Fig. 6. For the convolutional layer, it is composed of a series of convolutional kernels, each of which obtains a certain number of features from the input image to capture the hidden features and form a feature mapping. The convolutional layer can be represented as Eq. (1),

$$c_l^j = h(b_j + \sum_{m=1}^M w_m^j x_{l+m-1}^o) \quad (1)$$

where samples data input vector is $x_l^o = [x_1, x_2, \dots, x_n]$; l is the layer index; h is the activation function, a nonlinear operation of the CNN

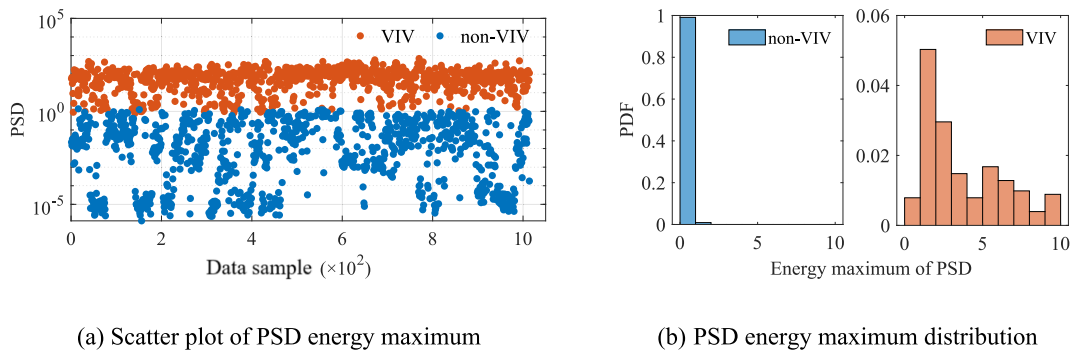


Fig.5. Statistical analysis of PSD energy maximum of suspenders vibration.

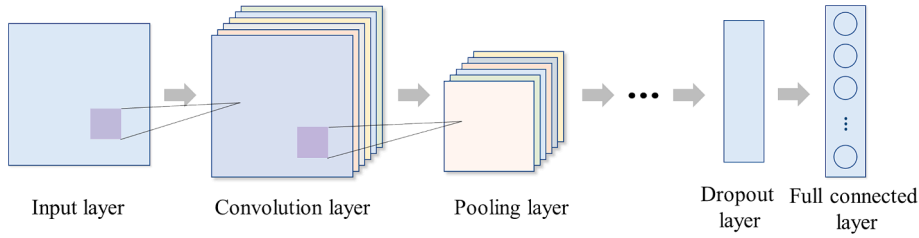


Fig.6. The typical CNN architecture.

network, which learns a nonlinear representation of the output volume of the previous layer of Conv by means of a nonlinear activation function such as the rectified linear unit (ReLU); b is the bias term for the j^{th} feature map; M is the kernel size, and w_m^j is the weight of the j^{th} feature map and m^{th} filter index.

The POOL layer can reduce the spatial dimensions of the extracted feature maps and the number of network parameters through some nonlinear numerical operations such as average pooling, sum pooling or maximum pooling. The maximum pooling (Max Pooling) is one of the most commonly used pooling methods, which is achieved by calculating the maximum value of a specified region in the feature map based on the equation.

The CNN in image data processing can effectively extract features such as texture, shape and edges of the images, which helps to classify and identify them [45]. It has been widely used in many image-based problems. A large number of applications of CNN in the field of SHM focus on vision-based surface defect detection [46,47] and image recognition for construction safety [48,49]. Inspired by them, this study employs CNNs to extract suspenders VIV time-domain features from vibration time-history images. The parameters of the convolution kernel (e.g., number, kernel size, step size, padding, etc.) are tuned and optimized according to the size of the input image and the network structure.

4.1.2. The BiLSTM network

LSTM-based networks are essentially extensions to RNN network that solve the gradient vanishing problem in a very concise way, allowing them to maintain long-term dependencies with the learning input [46,50,51]. The basic LSTM network generally consists of an input gate, an output gate and a forget gate (Fig. 7). The forget gate in the memory structure is controlled by a simple single-layer neural network. The activation of this gate is calculated by Eq. (2).

$$f_t = \sigma(W[x_t, h_{t-1}, C_{t-1}] + b_f) \quad (2)$$

where x_t is the input sequence; h_{t-1} is the output of the previous block; C_{t-1} is the memory of the previous LSTM block; b_f is the bias vector; and W is the logistic sigmoid function, representing a separate weight vector for each input.

The input gate decides whether to add new information to the LSTM memory. The gate consists of two layers, the sigmoid layer and the “tanh” layer: the sigmoid layer decides which values need to be updated, while the tanh layer creates a new vector of candidate values and adds them to the LSTM memory. The outputs of these two layers are computed by:

$$i_t = \sigma(W[x_t, h_{t-1}, C_{t-1}] + b_i) \quad (3)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh([x_t, h_{t-1}, C_{t-1}]) + b_c \quad (4)$$

The output gates are the parts of the LSTM module outputs. These outputs are calculated by Eq. (5) and Eq. (6), where O_t is the

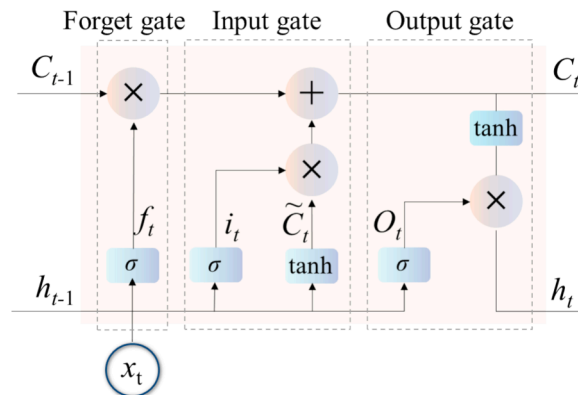


Fig.7. The flow chart of LSTM unit.

output value.

$$\sigma_t = \sigma(W[x_t, h_{t-1}, C_t] + b_o) \quad (5)$$

$$h_t = o_t \cdot \tanh(G_t) \quad (6)$$

BiLSTM is an extension of the LSTM network where two LSTMs are applied to the input data. In the first round, the LSTM is applied to the input sequence (i.e., the forward layer). In the second round, the reverse form of the input sequence is fed into the LSTM network (i.e., the backward layer) (Fig. 8). The BiLSTM is calculated as follows:

$$h_i = W_1 \cdot x_i + W_3 \cdot h_{i-1} + b_i \quad (7)$$

$$h'_i = W_2 \cdot x_i + W_4 \cdot h'_{i-1} + b'_i \quad (8)$$

$$y_i = h_i \oplus h'_i \quad (9)$$

where b_i and b'_i denote forward/backward LSTM bias vectors, respectively; and $\oplus$ denotes vector splicing.

The BiLSTM network has shown significant advantages in analyzing time series data for understanding long-term dependencies. It has been applied in structural health monitoring to feature extraction and representation learning [52], anomaly detection [53] and response prediction [54–56]. In this study, BiLSTM is used to capture the correlation between the PSDs of the suspender VIV for better identification of sequence features and also expected to improve the generalization of the framework.

4.1.3. MHA mechanism

The MHA mechanism is an effective method for extracting temporal correlation information and features among input data [52,57], which can ignore the time interval and compute the correlation information and features among input data in parallel (Fig. 9). When the sequence $Y = [y_1, y_2, \dots, y_d]$ is input, the query vector Q^i , the key-value vector K^i and the value vector V^i can be obtained.

$$Q^i = YW^{q,i} = [q_1^i, q_2^i, \dots, q_d^i] \quad (10)$$

$$K^i = YW^{k,i} = [k_1^i, k_2^i, \dots, k_d^i] \quad (11)$$

$$V^i = YW^{v,i} = [v_1^i, v_2^i, \dots, v_d^i] \quad (12)$$

The corresponding weight α can be obtained by Q^i and K^i , and then the weight α is normalized using the Softmax function to obtain the weight β .

$$\alpha_{ab}^i = \frac{q_a^i \cdot k_b^i}{\sqrt{d_k}} \quad (13)$$

$$\beta_{ab}^i = \text{Softmax}(\alpha_{ab}^i) \quad (14)$$

where $\sqrt{d_k}$ is a scaling factor to prevent the gradient loss when computing the Softmax function. Then the attention coefficient is calculated as Eq. (15).

$$\gamma_a^i = \sum_{b=1}^d \beta_{ab}^i \cdot v_b^i \quad (15)$$

The MHA mechanism is generally a linear transformation of the matrices Q , K , and V sequentially using different parameters, and the results of the linear transformation are input into the scaled dot product attention (Attention). The result of the calculation is

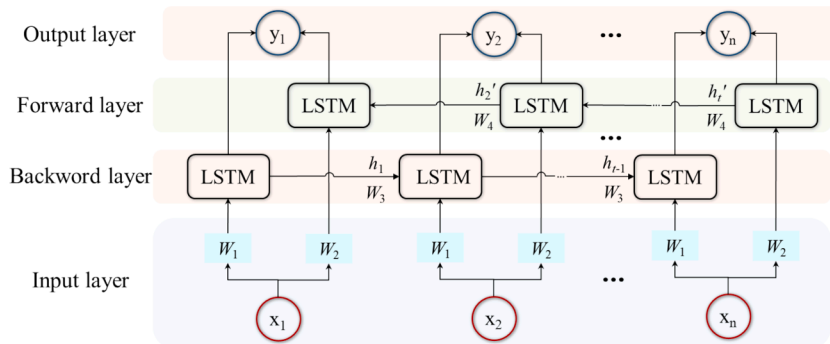


Fig.8. The typical BiLSTM network.

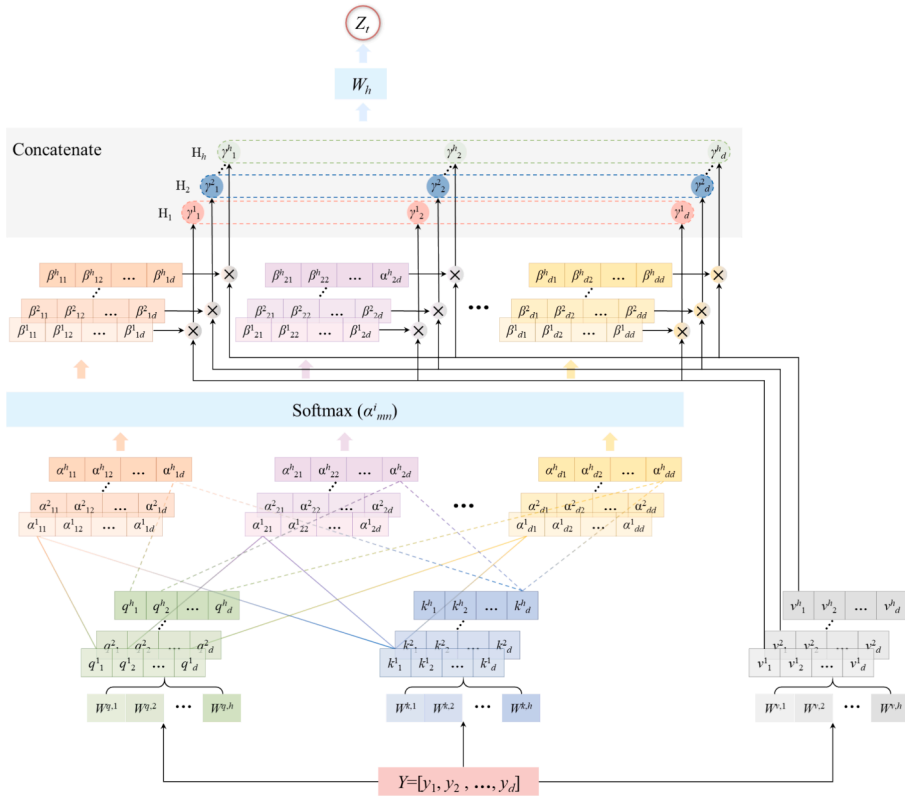


Fig.9. The MHA model.

denoted by H_i as shown in Eq. (16).

$$H_i = \text{Attention}(Q^i, K^i, V^i) = [\gamma_1^i, \gamma_2^i, \dots, \gamma_d^i] \quad (16)$$

The result of the calculation H_i is concatenated into a matrix and multiplied by the parameter W_h to complete the last linear

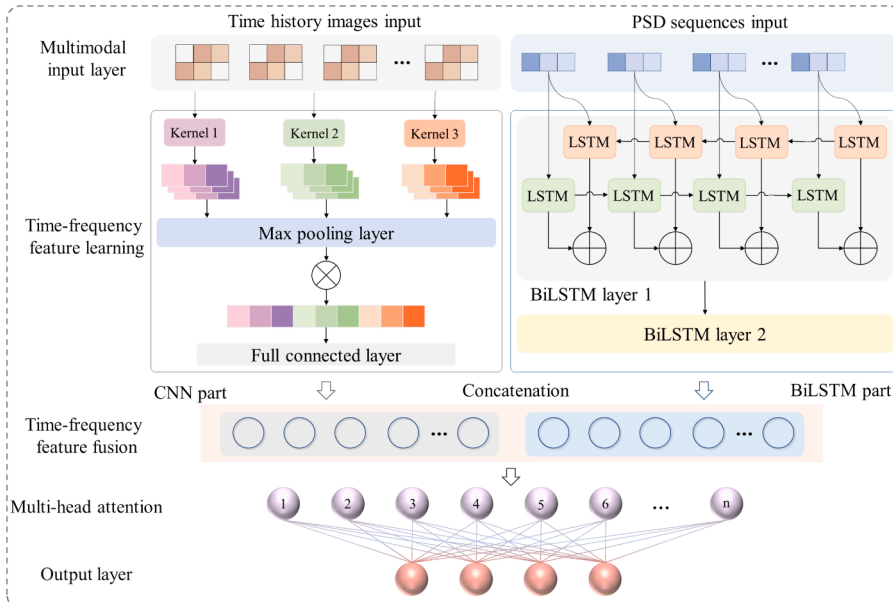


Fig.10. VIV identification framework based on CNN-BiLSTM-MHA.

transformation to get the operation result of the MHA mechanism, as shown in Eq. (17). h is the number of heads.

$$Z_t = \text{MultiHead}(Q, K, V) = \text{Concatenate}(H_1, H_2, \dots, H_h)W_h \quad (17)$$

In general, the MHA mechanism enables the network to better capture the dependencies between different locations in the input sequence, thus improving the performance and generalization of the network. By introducing multiple heads, the network can attend to different aspects of information in parallel, thus avoiding information loss or confusion and improving the efficiency of the network in utilizing different information. Thus, the MHA is exploited in structural health monitoring for multimodal data fusion, feature enhancement, and anomaly detection [56,58,59]. In this study, MHA is applied to the fusion of CNN and BiLSTM output information to efficiently capture the contribution degree of features extracted from vibration acceleration and PSD. These features extracted from the time domain and frequency domain can be focused and enhanced to improve the accuracy of suspender VIV identification.

4.2. Proposed framework

As mentioned in the previous sections, CNN network shows great potential in image processing, and BiLSTM network has advantages in processing abstract sequence data. The MHA mechanism utilizes multiple self-attentive layers that are independent of each other to correlate all the features and positional information of the input sequences and outputs more correlation information through the weighting to output more correlated information. In this study, a novel CNN-BiLSTM-MHA approach for automated identification of suspenders VIV is established by taking the information of two different modes, namely, vibration time-history images and PSD sequences, as inputs, respectively. The specific architecture of the method is presented in Fig. 10. As shown in Fig. 11, the recognition process of the proposed method can be summarized in the following stages, i.e. (I) data preparation; (II) network construction and training.

4.2.1. Data preparation

The monitoring data of three sets of suspenders from NQYRB are used as research data for this study. To denoise the measurement data, statistical data over a period of time rather than instantaneous measured data are used for modelling in this study. The statistical time interval is chosen to be 1-minute. Each event is accordingly represented as a time series consisting of 1-minute samples. Based on data anomaly detection and missing data reconstruction methods oriented towards structural health monitoring in previous studies [60–62], the required suspenders monitoring data are pre-processed. They were plotted as images using Python and Matplotlib and saved in image format for model input. After that, 3200 data are obtained from each of the three sets of suspenders for experimental validation. 2400 pieces of data are classified as training set and 800 data are classified as test set. In order to dynamically display the training process and prevent network overfitting, 20 % of the training data are used to validate performance at the end of each training

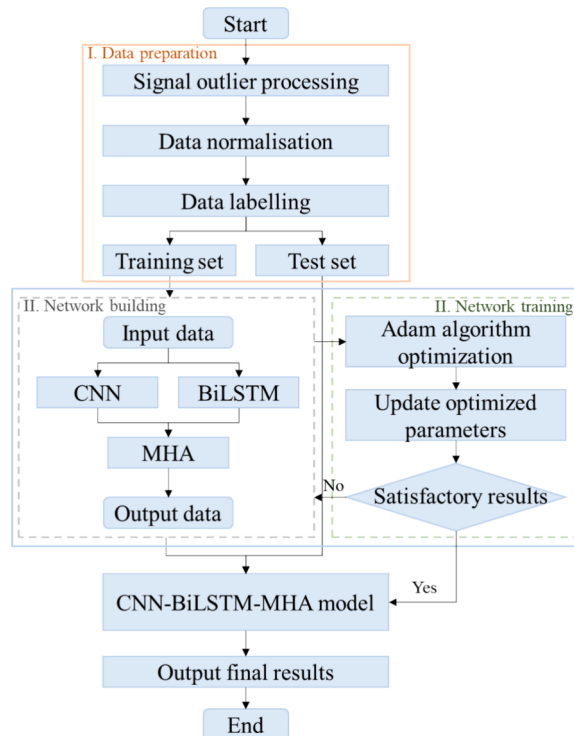


Fig.11. The identification process of CNN-BiLSTM-MHA framework.

epoch. The labelling of the datasets is performed according to the rules in Section 3.3, and each data set needs to contain only one labelled value principle.

4.2.2. Network building and training

The two inputs of the framework feed vibrational time history images and PSD sequences into two different networks, CNN and BiLSTM, respectively. The CNN network is adopted to process the images to address the non-stationary nature of the vibration time history data. The BiLSTM network receives the abstract sequence data from the vibration PSD and extracts the correlation information of the suspender VIV at different time intervals. The outputs of the two sub-architectures have the same shape and are merged into the inputs of the MHA mechanism. Afterwards the classification results of the suspenders VIV are output through the fully connected layer. The framework is parameterized as follows: Adam optimizer and sparse cross-entropy loss function are used with default parameter settings. The learning rate is also initially set to 0.001. Standard backpropagation method is used to train with a batch size of 32.

By fusing information from different modalities in this study, the algorithm can compensate for the respective limitations of 1D sequence signals and 2D images. For example, the images may be more sensitive to certain VIV stages, while the sequence signals are more sensitive to VIV or non-VIV. Multimodal fusion can help algorithms to better understand the essential features of the data, thereby reducing the risk of overfitting and improving the ability of algorithms to generalize on new data.

5. Framework feasibility and parametric analysis of VIV response

5.1. Identification results

To validate the effectiveness of the framework, the loss values and accuracy of training and validation are reported for three sets of suspenders, as shown in Fig. 12. In the first 100 epochs, the training loss value (orange lines) gradually decreases, and after 60 epochs, the training network basically reaches 97 % accuracy (blue lines). The confusion matrix obtained from the test data are shown in Fig. 13. In this study, the full developmental cycles of VIV are expressed separately as developmental stages of VIV (DV), stabilization stage of VIV (SV) and recessionary stage of VIV (RV). Overall, the framework seems to show satisfactory predictions for all three sets of suspenders, reaching more than 90 % for all of them. Among them, the number of DV and RV cases is relatively high. It is due to the close temporal proximity of DV and RV to the other stages and the closer proximity of VIV characteristics.

5.2. Results of the proposed framework versus the baseline network

To further validate the reasonability of the proposed framework in this study, four base models, CNN, LSTM, CNN-LSTM, CNN-LSTM-MHA, are selected for comparative analysis via the same three sets of suspender monitoring data. Among them, CNN and LSTM adopt only time history and PSD for unimodal information input, respectively, and the other networks adopt the same multimodal information input as the proposed framework. For a binary classification task, metrics such as precision, recall and F1 score are often used to evaluate the performance of the networks. For the multiclassification task in this study, the Macro-average method is chosen to evaluate the performance of the networks. The macro-averaged precision (MaP), macro-averaged recall (MaR) and macro-averaged F_1 score (MaF_1) are calculated as follows.

$$MaP = \frac{1}{n} \sum_{i=1}^n P_i = \frac{1}{n} \sum_{i=1}^n \frac{p_{ii}}{p_{i.}} \quad (18)$$

$$MaF_1 = \frac{1}{n} \sum_{i=1}^n F_{1i} = \frac{2}{n} \sum_{i=1}^n \frac{p_{ii}}{p_{i.} + p_{.i}} \quad (19)$$

$$MaR = \frac{1}{n} \sum_{i=1}^n R_i = \frac{1}{n} \sum_{i=1}^n \frac{p_{ii}}{p_{.i}} \quad (20)$$

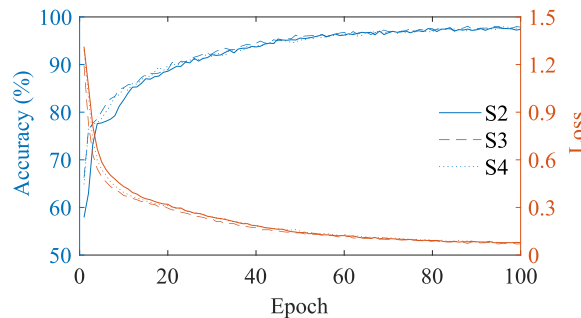


Fig.12. Accuracy and loss value curves for training and validation of the CNN-BiLSTM-MHA framework.

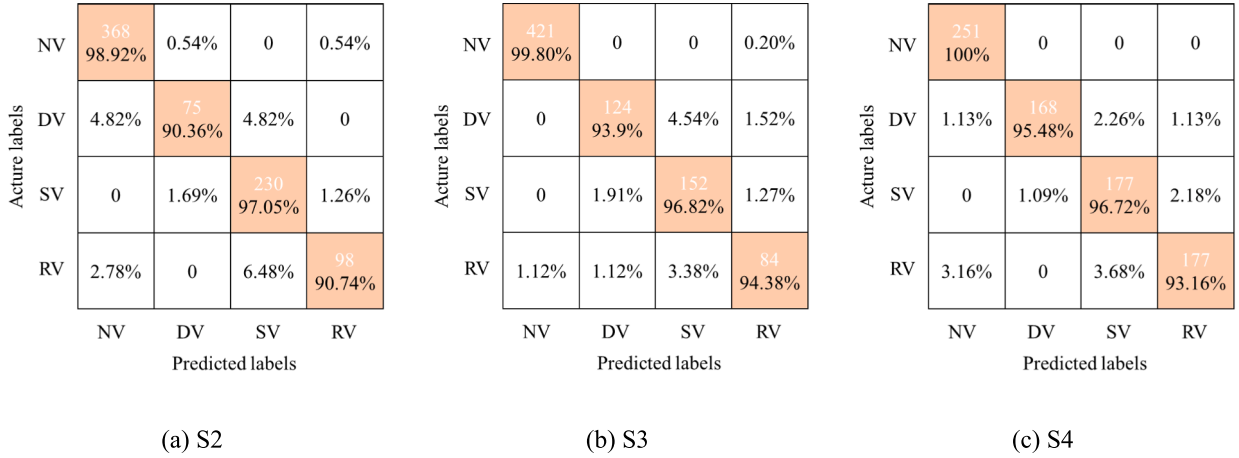


Fig.13. The confusion matrix obtained from the test data.

where TP is the i -th diagonal element; FP is the sum of the non-diagonal elements of the i -th row; and FN is the sum of the non-diagonal elements of the i -th column. Note that $TP_i = p_{ii}$, $TP_i + FP_i = p_i$, $TP_i + FN_i = p_i$.

The results of the performance evaluation metrics for the four base models as well as the proposed framework are given in Table 3. From the comparison results of the individual networks with the proposed framework, it can be seen that the three-evaluation metrics of the proposed multimodal classification framework of CNN-BiLSTM-MHA are better than the rest of the networks for the identification of three sets of suspenders, and the prediction performance has been significantly improved. The CNN network achieves unexpected results only by using time history images as input, but its performance still needs to be further improved. The LSTM network is difficult to identify the suspenders VIV only by using PSD sequences as input, and the model can be said to be a failure. The CNN-LSTM network and the CNN-LSTM-MHA network also achieved good identification results, but more experiments are needed to examine the generalization ability and applicability performance for different suspenders of different bridges. Overall, current model performs well on offline data, but may face challenges in processing latency and computational resources in real-time applications. In addition, identification errors in this study may arise from data imbalance or labelling errors. The model relies on a large amount of high-quality labelled data for training. In the process of data collection and labelling, the presence of data imbalance or labelling errors may affect the accuracy and generalization of the model.

5.3. Parameter analysis of suspenders VIV response

Typically, a complete VIV event lasts for several hours and consists of three stages (DV, SV, RV) [32]. To investigate the physical characteristics of VIV, parametric analyses are carried out for all three stages, including structural vibration response, time–frequency characteristics, modal damping ratio, wind velocity, and wind direction based on the identification results of the above framework. The results of the analyses are given here only for the S3.

5.3.1. DV

In the DV, the structure undergoes a process from initiation to stable vibration, during which the vibration characteristics may change. By investigating the developmental stage, the dynamic characteristics of the structure, including the parameters rules such as amplitude and frequency, can be revealed, providing a basis for subsequent optimal design and control. The characteristics and wind environment analysis results of the DV of the suspender are show in Fig. 14. The time–frequency diagram by short-time Fourier transform (STFT) for the period is depicted in Fig. 14(b). The colors represent the magnitude of PSD. Darker colors indicate higher PSD. It can be seen that the frequency in the VIV varies with time, which means that the VIV transforms in multiple modes rather than staying in a single mode. In the development stage, VIV occurs at a dominant frequency of 7.24 Hz. Since VIV occurs at relatively low

Table 3
Comparison of integrated framework identification results with baseline networks.

Model	S2			S3			S4		
	Pre (%)	Recall (%)	F1 score (%)	Pre (%)	Recall (%)	F1 score (%)	Pre (%)	Recall (%)	F1 score (%)
CNN-MHA	86.34	86.29	85.72	89.75	86.73	86.29	89.38	86.24	65.91
LSTM-MHA	72.16	30.27	24.35	78.21	35.42	25.36	79.14	40.49	37.28
CNN-LSTM	92.63	91.88	91.74	95.09	93.25	93.02	95.95	94.36	94.72
CNN-LSTM-MHA	93.67	91.59	92.82	96.25	93.47	93.52	95.86	94.37	93.82
Proposed model	95.84	93.12	94.07	97.03	94.53	95.69	97.45	95.62	95.84

frequencies, the amplitude of the vibrational acceleration is low and may not be obvious to the eyes, and therefore does not pose a driving safety concern. It does not mean that such VIV can be ignored, as low amplitude VIV may foreshadow the onset of high amplitude VIV. The results show that the method can identify VIV with insignificant vibration amplitude.

The modal damping ratios during VIV of the suspenders obtained using the automatic modal parameter identification method are displayed in Fig. 14(c) [27]. The markers of the same color are the damping ratios of the same modes. It can be found that the damping ratio of the suspenders at this stage decreases abruptly with the increase of acceleration RMS. When the acceleration RMS increases to 0.2 m/s^2 , it essentially drops below 0.2% , which is an important reason for the imminent occurrence of high-amplitude VIV. The damping ratio during VIV is determined based on the structural response to forced vibration. The suitability of this method for analyzing forced vibration data is uncertain. Therefore, the damping ratio results are for discussion only and not for interpretation. The wind characterization in the vicinity of the suspenders during this period is shown in Fig. 13(d). The wind velocity lies in the interval of $3 \text{ m/s} \sim 5.5 \text{ m/s}$ and the wind direction develop from nearly 30° to 60° , with a tendency to approach perpendicular to the direction of the suspender VIV, which further provides us with a warning.

5.3.2. SV

An in-depth study of VIV during the stabilization stage can help to better understand the vibration characteristics and response laws of the structure. It thus guides the optimal design of engineering structures and is available to managers for assessment, management and intervention. The vibration characteristics and wind environment of the SV of the suspender are shown in Fig. 15. At this stage, the most obvious VIV is observed visually, as shown by the acceleration time history in Fig. 15(a). At this stage, the dominant frequency of the suspender VIV changes from initially varying between 13 Hz and 15 Hz to mainly focusing on the dominant frequency of 15.15 Hz , indicating that the 21st order mode is the most severe vibration mode of the suspender VIV. Compared to the development stage, the damping ratio decreases further with increasing VIV and even drops below 0.1% when the acceleration RMS is greater than 20 m/s^2 . At this point the wind speed increased to $4 \text{ m/s} \sim 7 \text{ m/s}$ and the wind direction became more concentrated between 60° and 90° . From Fig. 15 and the identification results, it can be seen that the VIV of high amplitude vibration is not generated suddenly by the environmental excitation, but by the VIV with low amplitude vibration and low order frequency. Low frequency VIVs are easily initiated because lower modes are easily activated. This may be an interaction of vibration with vortex shedding, and the low-frequency VIV gradually develop into high-frequency VIV with high amplitude.

5.3.3. RV

The RV usually marks the gradual return of the structure to a steady state, where the system gradually consumes the original vibration energy and is no longer constantly subjected to external excitation. By investigating the decay phase, it is possible to find ways to reduce vibration energy loss and improve the operational efficiency of the structure. As can be seen from Fig. 16, the vibration amplitude decreases rapidly when VIV is about to disappear. The VIV with a frequency of 10 Hz is evident in Fig. 16(b), and the energy of multiple modes decreases gradually with decreasing acceleration amplitude. Obviously, during the recession process, the VIV

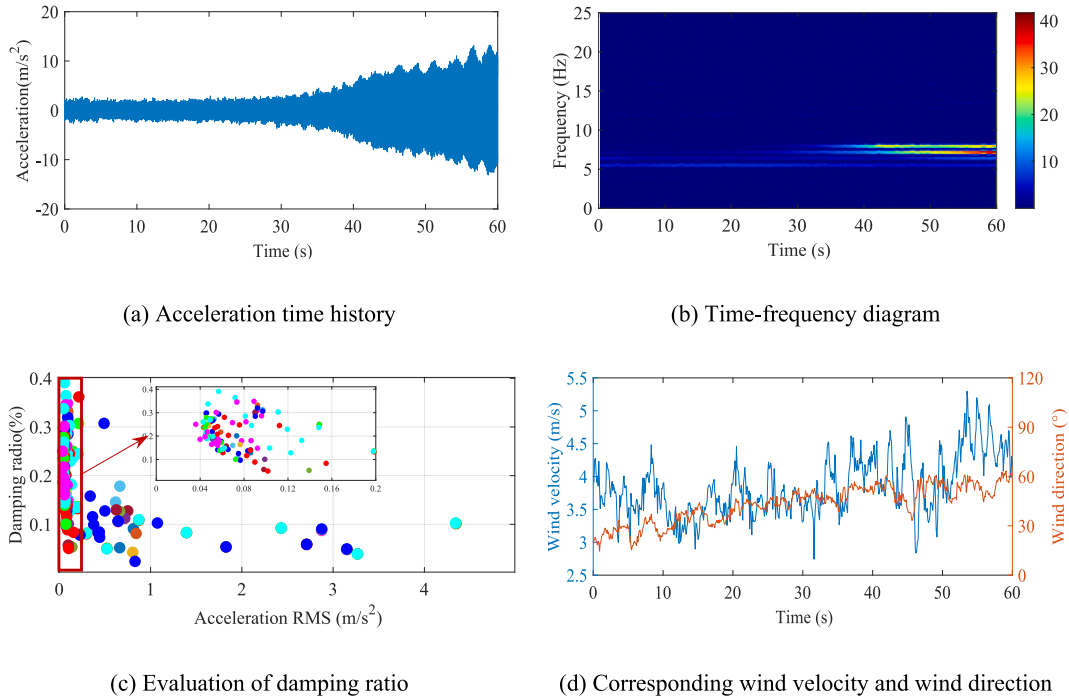


Fig.14. Characteristics and wind environment analysis of the DV of the suspender.

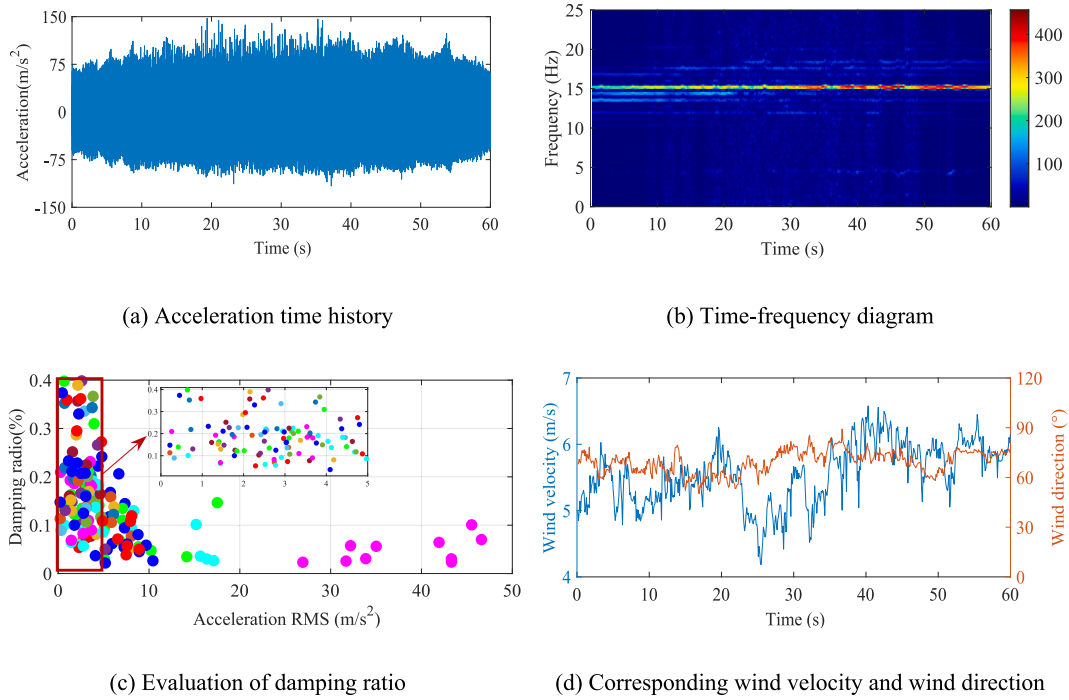


Fig.15. Characteristics and wind environment analysis of the SV of the suspender.

gradually shifts from high frequency to low frequency, and from high amplitude vibration to low amplitude vibration. The structural damping ratio has also increased at this stage as seen in Fig. 16(c). It is noted that the wind direction in Fig. 16(d) is expressed as $370^\circ - 400^\circ$ for the values between 0° and 40° for a better presentation. Overall, the suspender VIV is a process of accumulating and releasing energy, and it does not suddenly disappear. The method is capable of automatically identifying multi-stage suspenders VIV. Even for VIVs with small vibration amplitudes, the whole evolution of VIV from their beginning to their disappearance can be tracked. It shows the feasibility and novelty of the method in suspenders VIV identification.

6. Discussion

To verify the generalization of the suspender VIV identification approach in this study, it is transferred to Runyang Bridge based on transfer learning to check the performance of the model on different bridges. Runyang Bridge, opened to traffic on 30 April 2005, is a large bridge project group consisting of suspension bridges, cable-stayed bridges and prestressed concrete continuous girder bridges. The main span of its suspension bridge is 1490 m, which ranked the first in China and the third in the world when it was completed [63,64]. In this study, the trained suspenders VIV identification model is transferred to the monitoring data of a suspender of Runyang Bridge, and the main steps include: i) Data preprocessing and feature selection. To adapt to the input and feature selection of the model, the suspender monitoring data is pre-processed accordingly to ensure that it is consistent with the feature format and data range of the original model; ii) Model fine-tuning: some layers of the model are fine-tuned during the transfer process, including adjusting hyperparameters and optimizing model structure to improve its performance on Runyan Bridge data.

The performance of the model on the Runyon Bridge is evaluated through transfer learning. Fig. 17 shows the training curve of the transfer learning model. It can be observed that when the number of training rounds of the model reaches 100, the accuracy of the training and validation sets no longer increases and the model accuracy reaches 97.5 %. The results are close to those obtained on the NQYRB, indicating that the transfer learning approach is able to effectively transfer the knowledge of the model to a new dataset. In addition, the precision, recall and F1 score of the model are calculated, as shown in Table 4. All these metrics show the positive performance of the model on Runyang Bridge, which verifies the applicability of the model in the new environment.

Although transfer learning showed good results in this study, the generalization of the model remains an important issue. Structural differences and environmental changes were analysed on this issue. Differences in the layout, materials and other design aspects of the suspenders of Runyang Bridge and NQYRB may affect the performance of the model. By comparison, the adaptability of the model in dealing with different structures is analyzed. It is necessary to consider the introduction of environmental variables in future studies to test the performance of the model under different environmental conditions. To enhance the adaptability of the model under different bridge structures and environmental conditions, feature selection and engineering optimization needs to be carried out for the characteristics of different bridges to ensure that the model is able to handle diverse data. During the transfer process, the hyperparameters of the model are dynamically adjusted to adapt to the data characteristics of different bridges.

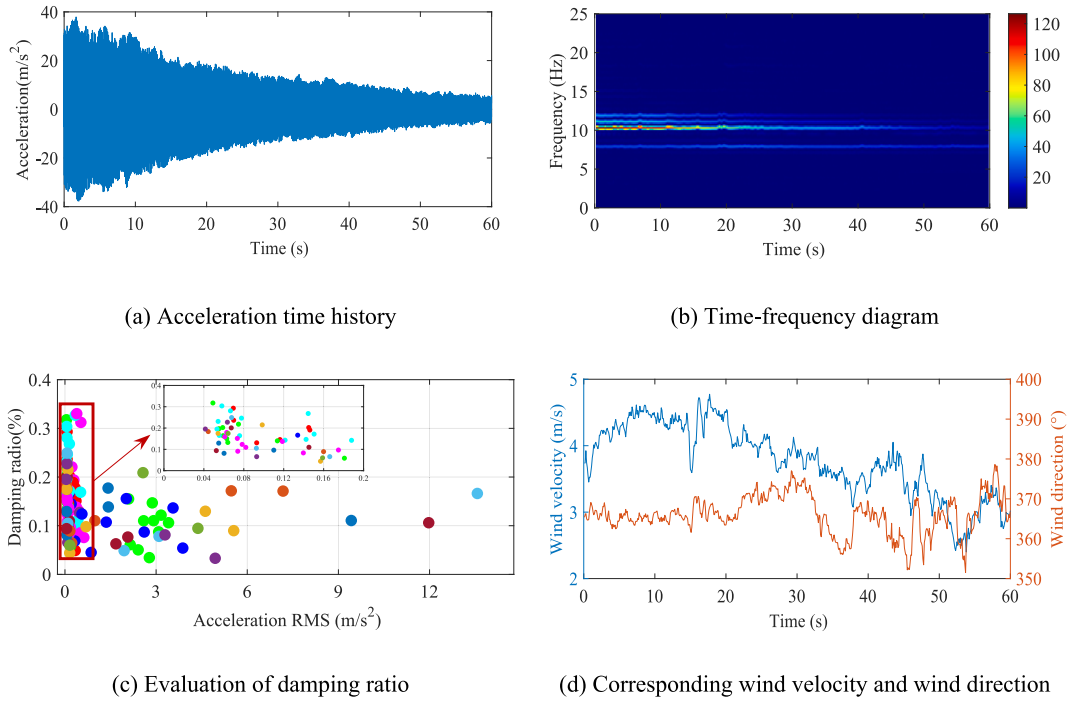


Fig.16. Characteristics and wind environment analysis of the RV of the suspender.

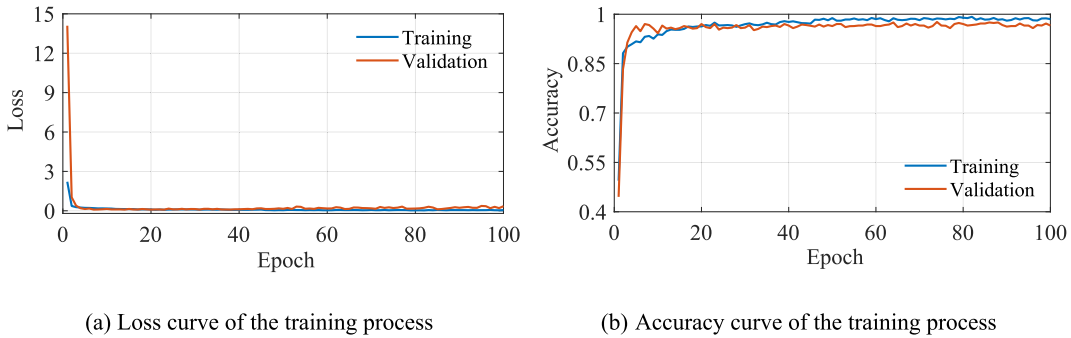


Fig.17. Transfer learning model training process curve.

Table 4

Performance metrics for the model prediction test set.

Categories	Precision	Recall	F1-Score	Support
NV	0.99	0.99	0.99	313
DV	0.95	0.96	0.96	215
SV	0.96	0.97	0.95	172
RV	0.97	0.96	0.97	246

7. Conclusion

In this study, a multimodal classification framework for automated identification of suspenders VIV is proposed. The framework extracts the information imbedded in both the time history and power spectral density (PSD) of the monitoring data and then fed them into several deep learning networks, i.e., CNN, BiLSTM and MHA, to judge the full process development of suspenders VIV. The validity of the proposed framework is verified using monitoring data from a full-scale kilometer-level long-span suspension bridge, namely, the NQYRB. The main conclusions can be summarized as follows.

- (1) Taking three sets of NQYRB suspenders with different diameters and lengths as the research objects, two vibration response parameters are determined as the indexes for the suspenders VIV identification through wind field and vibration characteristic parameter analyses based on big data significant difference method and distribution function fitting analysis.
- (2) The CNN-BiLSTM-MHA classification framework for VIV identification of suspenders in long-span suspension bridges was developed using multimodal information as input. The identification accuracy of the framework is higher than 95 % in all cases.
- (3) The trained framework was applied for parametric analysis of several randomly selected VIV events from the field monitoring database. The rationality of the framework was further verified by comparing it with several base models.

This study provides an effective analysis method and a reliable theoretical basis for grasping the whole process response of VIV of bridge suspenders, and filters effective data from massive monitoring data for vibration control. The methodology still has a lot of potential for development and advancement in the future, including:

- (1) By considering the effects of materials and geometries of different bridge suspenders, experiments can be carried out under more actual working conditions. This will contribute to a more thorough understanding of the performance of the VIV identification model.
- (2) Considering the application of the model to the SHM system for dynamic assessment and early warning of the bridge, the training and the inference efficiency of the model needs to be improved.
- (3) Explore how machine learning methods can be combined with physical models to better explain VIV phenomena. This includes using physical constraints to guide the training of machine learning models, or using model results to calibrate physical models.

CRedit authorship contribution statement

Xun Su: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Jianxiao Mao:** Writing – review & editing, Methodology, Funding acquisition. **Hao Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Hui Gao:** Writing – original draft, Validation. **Dan Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to gratefully acknowledge the support from the National Natural Science Foundation of China (No. 52338011, 52378291 and 52108274), the National Key Research and Development Program of China (No. 2022YFB3706704). And the first author gratefully acknowledges the financial support from the China Scholarship Council (CSC, 202406090201). and the SEU Innovation Capability Enhancement Plan for Doctoral Students (No. CXJH_SEU 24095).

Data availability

Data will be made available on request.

References

- [1] X.G. Hua, Z.Q. Chen, X. Lei, et al., Monitoring and control of wind-induced vibrations of hanger ropes of a suspension bridge, *Smart Struct. Syst. Int. J.* 23 (6) (2019) 683–693.
- [2] W. Chen, D. Gao, H. Li, et al., Wake-flow-induced vibrations of vertical hangers behind the tower of a long-span suspension bridge, *Eng. Struct.* 169 (2018) 188–200.
- [3] Z. Xu, H. Wang, C. Xing, et al., Physics guided wavelet convolutional neural network for wind-induced vibration modeling with application to structural dynamic reliability analysis, *Eng. Struct.* 297 (2023) 117027.
- [4] X. Wu, F. Ge, Y. Hong, A review of recent studies on vortex-induced vibrations of long slender cylinders, *Journal of Fluids and Structures* 28 (2012) 292–308.
- [5] W.L. Chen, Q.Q. Zhang, H. Li, et al., An experimental investigation on vortex induced vibration of a flexible inclined cable under a shear flow, *J. Fluids Struct.* 54 (2015) 297–311.
- [6] K.H. Halse, On Vortex Shedding and Prediction of Vortex-Induced Vibrations of Circular Cylinders, Norwegian University of Science and Technology, 1997.
- [7] Y. Wang, X. Chen, Extraction of aerodynamic damping and prediction of vortex-induced vibration of bridge deck using CFD simulation of forced vibration, *J. Wind Eng. Ind. Aerodyn.* 224 (2022) 104982.
- [8] H. Gao, H. Wang, J. Mao, et al., Field monitoring and control for vortex-induced vibration of hanger cables in suspension bridge, *J. Wind Eng. Ind. Aerodyn.* 248 (2024) 105712.
- [9] B. Wang, S. Hao, Y.L. Xu, et al., Modelling of vortex-induced force and prediction of vortex-induced vibration of a bridge deck using method of multiple scales, *J. Wind Eng. Ind. Aerodyn.* 241 (2023) 105533.
- [10] M.J. Thorsen, S. Sævik, C.M. Larsen, A simplified method for time domain simulation of cross-flow vortex-induced vibrations, *J. Fluids Struct.* 49 (2014) 135–148.
- [11] M. Zhang, S. Fu, H. Ren, et al., A hybrid FEM-DNN-based vortex-induced vibration prediction method for flexible pipes under oscillatory flow in the time domain, *Ocean Eng.* 246 (2022) 110488.
- [12] C. Hu, L. Zhao, Y. Ge, Time-frequency evolutionary characteristics of aerodynamic forces around a streamlined closed-box girder during vortex-induced vibration, *J. Wind Eng. Ind. Aerodyn.* 182 (2018) 330–343.

- [13] Z. Kang, C. Zhang, R. Chang, et al., A numerical investigation of the effects of Reynolds number on vortex-induced vibration of the cylinders with different mass ratios and frequency ratios, *Int. J. Naval Architect. Ocean Eng.* 11 (2) (2019) 835–850.
- [14] Y. Song, Z. Ti, Y. Li, An efficient two-stage hybrid framework to evaluate vortex-induced vibration for bridge deck based on divergent vibration, *J. Wind Eng. Ind. Aerodyn.* 233 (2023) 105316.
- [15] B. Ge, R. Ma, F. Li, et al., Probabilistic vortex-induced vibration occurrence prediction of the twin-box girder for long-span cable-stayed bridges based on wind tunnel tests, *Eng. Struct.* 262 (2022) 114325.
- [16] M.F. Huang, B.Y. Zhang, Y. Guo, et al., Prediction and suppression of vortex-induced vibration for steel tubes with bolted joints in tubular transmission towers, *J. Struct. Eng.* 147 (9) (2021) 04021128.
- [17] M. Zhang, Y. Song, A. Abdelkefi, et al., Vortex-induced vibration of a circular cylinder with nonlinear stiffness: prediction using forced vibration data, *Nonlinear Dyn.* 108 (3) (2022) 1867–1884.
- [18] X.G. Hua, Z.Q. Chen, W. Chen, et al., Investigation on the effect of vibration frequency on vortex-induced vibrations by section model tests, *Wind Struct.* 20 (2) (2015) 349–361.
- [19] Lu Z, Fu S, Zhang M, et al. A Non-Iterative Method for Vortex Induced Vibration Prediction of Marine Risers[C]. *International Conference on Offshore Mechanics and Arctic Engineering*. American Society of Mechanical Engineers, 2017, 57649: V002T08A028.
- [20] M. Zhang, F. Xu, H. Yu, A simplified model to evaluate peak amplitude for vertical vortex-induced vibration of bridge decks, *Int. J. Mech. Sci.* 192 (2021) 106145.
- [21] C. Gao, Z. Shi, W. Zhang, Data-knowledge-driven semi-empirical model augmentation method for nonlinear vortex-induced vibration, *Nonlinear Dyn.* (2023) 1–26.
- [22] Y. Bao, H.B. Zhu, P. Huan, et al., Numerical prediction of vortex-induced vibration of flexible riser with thick strip method, *J. Fluids Struct.* 89 (2019) 166–173.
- [23] Ma B, Srinil N. Numerical Prediction of 3-D Vortex-Induced Vibration of Catenary Riser in Planar and Non-Planar Flows[C]. *International Conference on Offshore Mechanics and Arctic Engineering*. American Society of Mechanical Engineers, 2021, 85185: V008T08A020.
- [24] D. Deng, W. Zhao, D. Wan, Vortex-induced vibration prediction of a flexible cylinder by three-dimensional strip model, *Ocean Eng.* 205 (2020) 107318.
- [25] E. Dobrucali, O.K. Kinaci, URANS-based prediction of vortex induced vibrations of circular cylinders, *Journal of Applied Fluid Mechanics* 10 (3) (2017) 957–970.
- [26] J. Park, S. Kim, H.K. Kim, Effect of gap distance on vortex-induced vibration in two parallel cable-stayed bridges, *Journal of Wind Engineering and Industrial Aerodynamics* 162 (2017) 35–44.
- [27] J. Mao, X. Su, H. Wang, et al., Automated Bayesian operational modal analysis of the long-span bridge using machine-learning algorithms, *Eng. Struct.* 289 (2023) 116336.
- [28] J. Lim, S. Kim, H.K. Kim, Using supervised learning techniques to automatically classify vortex-induced vibration in long-span bridges, *J. Wind Eng. Ind. Aerodyn.* 221 (2022) 104904.
- [29] S. Li, S. Laima, H. Li, Cluster analysis of winds and wind-induced vibrations on a long-span bridge based on long-term field monitoring data, *Engineering Structures* 138 (2017) 245–259.
- [30] S. Kim, T. Kim, Machine-learning-based prediction of vortex-induced vibration in long-span bridges using limited information, *Eng. Struct.* 266 (2022) 114551.
- [31] D. Dan, H. Li, Monitoring, intelligent perception, and early warning of vortex-induced vibration of suspension bridge, *Structural Control and Health Monitoring* 29 (5) (2022) e2928.
- [32] S. Li, S. Laima, H. Li, Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression, *J. Wind Eng. Ind. Aerodyn.* 172 (2018) 196–211.
- [33] M. He, P. Liang, Y. Zhang, et al., Identification, tracking and warning of vortex induced vibration using k-means clustering method, *Struct. Infrastruct. Eng.* (2022) 1–14.
- [34] M. Arul, A. Kareem, D.K. Kwon, Identification of vortex-induced vibration of tall building pinnacle using cluster analysis for fatigue evaluation: Application to Burj Khalifa, *J. Struct. Eng.* 146 (11) (2020) 04020234.
- [35] H.W. Zhao, Y.L. Ding, A.Q. Li, et al., Evaluation and early warning of vortex-induced vibration of existed long-span suspension bridge using multisource monitoring data, *J. Perform. Constr. Facil* 35 (3) (2021) 04021007.
- [36] S. Xu, R. Ma, D. Wang, et al., Prediction analysis of vortex-induced vibration of long-span suspension bridge based on monitoring data, *J. Wind Eng. Ind. Aerodyn.* 191 (2019) 312–324.
- [37] W. Xu, Z. He, L. Zhai, et al., Vortex-induced vibration prediction of an inclined flexible cylinder based on machine learning methods, *Ocean Eng.* 282 (2023) 114956.
- [38] W. Xu, K. Jia, Y. Ma, et al., Multispan classification methods and interaction mechanism of submarine pipelines undergoing vortex-induced vibration, *Appl. Ocean Res.* 120 (2022) 103027.
- [39] Z. Huang, Y. Li, X. Hua, et al., Automatic identification of bridge vortex-induced vibration using random decrement method, *Appl. Sci.* 9 (10) (2019) 2049.
- [40] Z. Wang, J. Zhu, Z. Zhang, Machine learning-based deep data mining and prediction of vortex-induced vibration of circular cylinders, *Ocean Eng.* 285 (2023) 115313.
- [41] Z. Ma, J. Li, W. Feng, et al., Prediction of Vortex-Induced Vibration Response of Open Bridge Girder Based on Machine Learning Method, *Int. J. Struct. Stab. Dyn.* (2023).
- [42] Z. Lu, S. Fu, M. Zhang, et al., An efficient time-domain prediction model for vortex-induced vibration of flexible risers under unsteady flows, *Mar. Struct.* 64 (2019) 492–519.
- [43] B. Zhang, W. Qiu, Improving time-domain prediction of vortex-induced vibration of marine risers, *Marine Systems & Ocean Technology* 13 (2018) 13–25.
- [44] G. Lu, Y. Liu, J. Wang, et al., CNN-BiLSTM-Attention: A multi-label neural classifier for short texts with a small set of labels, *Inf. Process. Manag.* 60 (3) (2023) 103320.
- [45] C. Chen, Z. Hua, R. Zhang, et al., Automated arrhythmia classification based on a combination network of CNN and LSTM, *Biomedical Signal Processing and Control* 57 (2020) 101819.
- [46] S.A.V. Shajihan, S. Wang, G. Zhai, et al., CNN based data anomaly detection using multi-channel imagery for structural health monitoring, *Smart Struct Syst* 29 (1) (2022) 181–193.
- [47] Z. Tang, Z. Chen, Y. Bao, et al., Convolutional neural network-based data anomaly detection method using multiple information for structural health monitoring, *Struct. Control Health Monit.* 26 (1) (2019) e2296.
- [48] X.W. Ye, T. Jin, C.B. Yun, A review on deep learning-based structural health monitoring of civil infrastructures, *Smart Struct. Syst* 24 (5) (2019) 567–585.
- [49] Q. Fang, H. Li, X. Luo, et al., Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, *Autom. Constr.* 85 (2018) 1–9.
- [50] S. Siarni-Namini, N. Tavakoli, A.S. Namin. The performance of LSTM and BiLSTM in forecasting time series[C]. *2019 IEEE International conference on big data (Big Data)*. IEEE, 2019: 3285–3292.
- [51] W. Lu, J. Li, J. Wang, et al., A CNN-BiLSTM-AM method for stock price prediction, *Neural Comput. Applic.* 33 (10) (2021) 4741–4753.
- [52] Z. Wang, Z. Peng, Structural acceleration response reconstruction based on BiLSTM network and multi-head attention mechanism, *Structures* 64 (2024) 106602.
- [53] G. Qu, Y. Xia, L. Sun, et al., Behavior Expectation-Based Anomaly Detection in Bridge Deflection Using AOA-BiLSTM-TPA: Considering Temperature and Traffic-Induced Temporal Patterns, *Struct. Control Health Monit.* 2024 (1) (2024) 2337057.
- [54] Z. Ma, Y. Sun, H. Ji, et al., A CNN-BiLSTM-Attention approach for EHA degradation prediction based on time-series generative adversarial network, *Mech. Syst. Sig. Process.* 215 (2024) 111443.
- [55] H. Ju, H. Shi, W. Shen, et al., An accurate and low-cost vehicle-induced deflection prediction framework for long-span bridges using deep learning and monitoring data, *Eng. Struct.* 310 (2024) 118094.
- [56] Y. Shen, B. Tang, B. Li, et al., Remaining useful life prediction of rolling bearing based on multi-head attention embedded Bi-LSTM network, *Measurement* 202 (2022) 111803.

- [57] D. Qiu, B. Yang, Text summarization based on multi-head self-attention mechanism and pointer network, *Complex & Intelligent Systems* (2022) 1–13.
- [58] Q. Liu, J. Wang, H. Dai, et al., Bridge Structural Damage Identification Based on Parallel Multi-head Self-attention Mechanism and Bidirectional Long and Short-term Memory Network, *Arab. J. Sci. Eng.* (2024) 1–13.
- [59] S. Ghazimoghadam, S.A.A. Hosseinzadeh, A novel unsupervised deep learning approach for vibration-based damage diagnosis using a multi-head self-attention LSTM autoencoder, *Measurement* 229 (2024) 114410.
- [60] Y.M. Zhang, H. Wang, Y. Bai, et al., Bayesian dynamic regression for reconstructing missing data in structural health monitoring, *Struct. Health Monit.* 21 (5) (2022), 147592172110537.
- [61] Y.M. Zhang, H. Wang, H.P. Wan, et al., Anomaly detection of structural health monitoring data using the maximum likelihood estimation-based Bayesian dynamic linear model, *Struct. Health Monit.* 20 (6) (2021) 2936–2952.
- [62] J. Mao, H. Wang, B.F. Spencer Jr, Toward data anomaly detection for automated structural health monitoring: Exploiting generative adversarial nets and autoencoders, *Struct. Health Monit.* 20 (4) (2021) 1609–1626.
- [63] H. Wang, A. Li, T. Guo, et al., Establishment and application of the wind and structural health monitoring system for the Runyang Yangtze River Bridge, *Shock Vib.* 2014 (1) (2014) 421038.
- [64] H. Wang, T. Guo, T.Y. Tao, et al., Study on wind characteristics of Runyang suspension bridge based on long-term monitored data, *Int. J. Struct. Stab. Dyn.* 16 (04) (2016) 1640019.